

Linear Algebra Exam 3 Study Guide

1 Section 5.2: Inner Product Spaces

- An **inner product** (denoted as $\langle \vec{u}, \vec{v} \rangle$) is literally just any operation that satisfies the following axioms (let $\vec{u}, \vec{v}$, & $\vec{w}$ be in a vector space V , and $c \in \mathbb{R}$):

- $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$
- $\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$
- $c\langle \vec{u}, \vec{v} \rangle = \langle c\vec{u}, \vec{v} \rangle = \langle \vec{u}, c\vec{v} \rangle$
- $\langle \vec{u}, \vec{u} \rangle \geq 0$ & $\langle \vec{u}, \vec{u} \rangle = 0$ iff $\vec{u} = 0$

Note: The dot product is an example of an inner product (it is also called the Euclidean inner product)

- Note: An inner product can also be done on matrices, and even definite integrals
- Inner product space:** a vector space with an inner product defined
- Length of a vector:** $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$
- Distance between 2 vectors:** $d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$
- The angle between 2 non-zero vectors:**

$$\cos(\theta) = \frac{\langle \vec{u}, \vec{v} \rangle}{\|\vec{u}\| \|\vec{v}\|}, \text{ \& } 0 \leq \theta \leq \pi$$

- Orthogonal:** $\langle \vec{u}, \vec{v} \rangle = 0$
- Unit vector eq:** $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$ (or to normalize)
- Orthogonal Projections:** a vector broken down into 2 parts
 - $\vec{u} = \hat{u} + \vec{z}$
 - * $\vec{u}$: a vector
 - * $\hat{u}$: the projection of $\vec{u}$ on another vector $\vec{v}$. Think of it has the shadow that $\vec{u}$ casts on all scalar multiples of $\vec{v}$
 - * $\vec{z}$: The vector orthogonal to $\vec{v}$
 - $\hat{u} = \text{proj}_{\vec{v}} \vec{u} = \frac{\langle \vec{u}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}$
 - $\vec{z} = \vec{u} - \hat{u}$

2 Section 5.3: Orthogonal Basis - Gram-Schmidt

- Vector spaces can have many basis, like how $\mathbb{R}^3$ has the basis $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ (the standard basis for $\mathbb{R}^3$). Standard basis have some important characteristics, like how all the vectors in it are mutually orthogonal

$$(1, 0, 0) \cdot (0, 1, 0) = 0$$

$$(1, 0, 0) \cdot (0, 0, 1) = 0$$

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- Orthogonal set:** $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \in \mathbb{R}^n$ such that $\langle \vec{u}_i, \vec{u}_j \rangle = 0 \forall i \neq j$. Basically, all vectors are orthogonal to each other
- Othonormal set:** $S = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \in \mathbb{R}^n$ such that: $\langle \vec{u}_i, \vec{u}_j \rangle = 0 \forall i \neq j$, and every vector is a unit vector.

- **Gram-Schmidt Orthonormalization Algorithm:** Algorithm to find orthonormal bases, where you start with a basis, you turn it into an orthonormal basis, then normalize each vector.

1. Let $B = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ be a basis for an inner product space V
2. Let $B' = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}$, where

$$\begin{aligned}\vec{w}_1 &= \vec{v}_1 \\ \vec{w}_2 &= \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 \\ \vec{w}_3 &= \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \frac{\langle \vec{v}_3, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2 \\ &\vdots \\ \vec{w}_n &= \vec{v}_n - \frac{\langle \vec{v}_n, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle} \vec{w}_1 - \frac{\langle \vec{v}_n, \vec{w}_2 \rangle}{\langle \vec{w}_2, \vec{w}_2 \rangle} \vec{w}_2 - \dots - \frac{\langle \vec{v}_n, \vec{w}_{n-1} \rangle}{\langle \vec{w}_{n-1}, \vec{w}_{n-1} \rangle} \vec{w}_{n-1}\end{aligned}$$

Then B' is an orthonormal basis for V

3. Let $\vec{u}_i = \frac{\vec{w}_i}{\|\vec{w}_i\|}$. Then $B'' = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ is an orthonormal basis for V

Also, $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} = \text{span}\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n\}$ for $k = 1, 2, \dots, n$

3 Section 6.1: Introductions to Linear Transformations

- **Linear Transformations:** Denoted as $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and is a function that satisfies for $\vec{u}, \vec{v} \in \mathbb{R}^n$ and $k \in \mathbb{R}$ the following:
 - The additive property: $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ (or addition)
 - The homogeneity property: $T(k\vec{u}) = kT(\vec{u})$ (or scalar multiplication)
 - If there is an $A_{m \times n}$ matrix that applies the linear transformation through matrix multiplication.
 - * $T(\vec{x}) = A_{m \times n} \vec{x}_{n \times 1}$
 - * If $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, then $A_{n \times n}$
 - $T(-\vec{v}) = -T(\vec{v})$
 - $T(0) = 0$
- **Zero transformation:** $T(\vec{v}) = 0$, for all $\vec{v}$
- **Identity transformation:** $T(\vec{v}) = \vec{v}$, for all $\vec{v}$

4 Section 6.2: The Kernel and Range of Linear Transformation

- **Kernel:** all vectors $\vec{v}$ in V such that that satisfy $T(\vec{v}) = 0$ where $T : V \rightarrow W$. Denoted by $\ker(T)$
 - Looking at $T(\vec{x}) = A\vec{x}$, with the kernel would be all $\vec{x}$ where $A\vec{x} = 0$. This is also the nullspace of A
 - If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, kernel is in $\mathbb{R}^n$
- **Range:** of a linear transformation $T : V \rightarrow W$ is a subspace of W . Think of it as the range of a function, where the range is a subset of the codomain
 - Looking at $T(\vec{x}) = A\vec{x}$, the range would be the column space of A
 - If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, range is in $\mathbb{R}^m$
- **The dimension theorem:** says that $\text{rank}(T) + \text{kern}(T) = \text{dimension of the domain}$

- $\text{rank}(T)$ is the dimension of the row space or column space of T , or how many vectors are in the row space or column space
- **One-to-one (or bijective):** When $T(\vec{v}_1) = T(\vec{v}_2)$ iff $\vec{v}_1 = \vec{v}_2$ (same as one-to-one in a function)
 - A transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is one-to-one iff $\ker(T) = \vec{0}_n$
 - * If $R = \text{rref}(A)$ has non-pivot columns (free variables) then T is not one-to-one
- **Onto transformation:** When $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ range is all of $\mathbb{R}^m$ (same as onto in a function)
 - Onto iff $\text{rank}(T) = m$
 - When $R = \text{rref}(A)$ has at least one row consisting all zeros, then not onto

5 Section 6.3: Matrices for Linear Transformations

- Example: 2 representations of $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$
 - $T(x_1, x_2, x_3) = (2x_1 + x_2 - x_3, -x_1 + 3x_2 - 2x_3, 3x_2 + 4x_3)$
 - $T(\vec{x}) = A\vec{x} = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 3 & -2 \\ 0 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

The second is simpler to write and understand

- **Matrix addition/subtraction:** Given $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$, $A + B = [a_{ij} + b_{ij}]$ and $1 \leq i \leq m$ and $1 \leq j \leq n$
- **Recall:** $f \circ g = f(g(x))$
 - The same can be done with transformations: $T_1 : \mathbb{R}^n \rightarrow \mathbb{R}^k$, $T_2 : \mathbb{R}^k \rightarrow \mathbb{R}^m$, $T_2 \circ T_1 : \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$\vec{u} \in \mathbb{R}^n$$

$$T_1(\vec{u}) = \vec{v} \in \mathbb{R}^k$$

$$T_2(\vec{v}) = \vec{w} \in \mathbb{R}^m$$

$$(T_2 \circ T_1)(\vec{u}) = T_2(T_1(\vec{u})) = T_2(\vec{v}) = \vec{w}$$

- **Inverse Linear Transformation:** If $T_1 : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $T_2 : \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations such that for every $\vec{v}$ in $\mathbb{R}^n$,

$$T_2(T_1(\vec{v})) = \vec{v}$$

and

$$T_1(T_2(\vec{v})) = \vec{v}$$

then T_2 is the **inverse** of T_1 , and T_1 is **invertible**

6 Section 7.1: Eigenvalues and Eigenvectors

- **Eigenvalue problem:** When A is an $n \times n$ matrix, do nonzero vectors $\vec{x}$ in $\mathbb{R}^n$ exist such that $A\vec{x}$ is a scalar multiple of $\vec{x}$, or $A\vec{x} = \lambda\vec{x}$?
 - **Eigenvalue:** scalar denoted as λ
 - **Eigenvectors:** The nonzero vector $\vec{x}$ and cannot be zero
- **Eigenspaces:** The set of all eigenvectors of a given eigenvalue λ , together with the $\vec{0}$. Since each eigenvalue can have infinitely many eigenvectors. This is also a subspace of $\mathbb{R}^n$.
- **Finding eigenvalue of A:** a scalar such that $\det(\lambda I - A) = 0$

- $|\lambda I - A| = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_2\lambda^2 + c_1\lambda + c_0$
- Finding the roots of the above give you the eigenvalues of A
- **Finding eigenvectors of λ :** The nonzero solutions of $(A - \lambda I)\vec{x} = 0$
 - For each λ , find the eigenvectors by solving the above equation. Requires row reducing, and there must be at least one row of zeros
- **Eigenvalues of Triangular Matrices:** If A is an $n \times n$ triangular matrix, then its eigenvalues are the entries on its main diagonal

7 Section 7.2: Diagonalization

- **Diagonalization problem:** For a square matrix A , does there exist an invertible matrix P such that $P^{-1}AP$ is diagonal?
- **Square matrix:** Diagonalizable when A is similar to a diagonal matrix D . Or when there is an invertible matrix P such that $D = P^{-1}AP$.
 - A matrix A_n is diagonal if it has n linearly independent eigenvectors.
 - Similar matrices have the same eigenvalues
- **Steps for Diagonalizing a Square Matrix:** Let A be a square matrix
 - Find n linearly independent eigenvectors $\vec{p}_1, \vec{p}_2, \dots, \vec{p}_n$ for A (if possible) with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. If n linearly independent vectors don't exist, then A is not diagonalizable
 - Let P be the $n \times n$ matrix whose columns consist of these eigenvectors. That is $P = [\vec{p}_1 \ \vec{p}_2 \ \dots \ \vec{p}_n]$
 - The diagonal matrix $D = P^{-1}AP$ will have the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ on its main diagonal.

8 Section 7.3: Symmetric Matrices and Orthogonal Diagonalization

- **Symmetric matrix:** When a square matrix A is equal to its transpose, $A = A^T$. Guaranteed to be diagonalizable
- **Symmetric matrix properties:**
 1. A is diagonalizable
 2. All eigenvalues of A are real
 3. If λ is an eigenvalue of A with multiplicity k , then λ has k linearly independent eigenvectors (or that the eigenspace of λ has dimension k)
 4. If λ_1 and λ_2 are distinct eigenvalues of A , then their corresponding eigenvectors $\vec{x}_1$ and $\vec{x}_2$ are orthogonal
 5. A is orthogonally diagonalizable (and has real eigenvalues) iff A is symmetric
- **Orthogonal matrices:** When a square matrix P is invertible, and $P^{-1} = P^T$. Also, its column vectors form an orthonormal set
- **Orthogonal Diagonalization of a Symmetric Matrix:** Let A be an $n \times n$ symmetric matrix.
 1. Find all eigenvalues of A and determine the multiplicity of each.
 2. For each eigenvalue of multiplicity 1, find a unit eigenvector. (Find any eigenvector and then normalize it.)
 3. For each eigenvalue of multiplicity of $k \geq 2$, find a set of k linearly independent eigenvectors. If this set is not orthonormal, then apply the Gram-Schmidt orthonormalization process
 4. The results from 2 and 3 produce an orthonormal set of n eigenvectors. Use these to form the columns of P . The matrix $P^{-1}AP = P^TAP = D$ will be diagonal.